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GEOAI FUSION 2026

Module 4 Session 3: Infrastructure Planning with GeoAI

AGAIF 2026 Boot Camp 1
Foundation of GeoAI

Thursday 9 July 2026

Delivered by:
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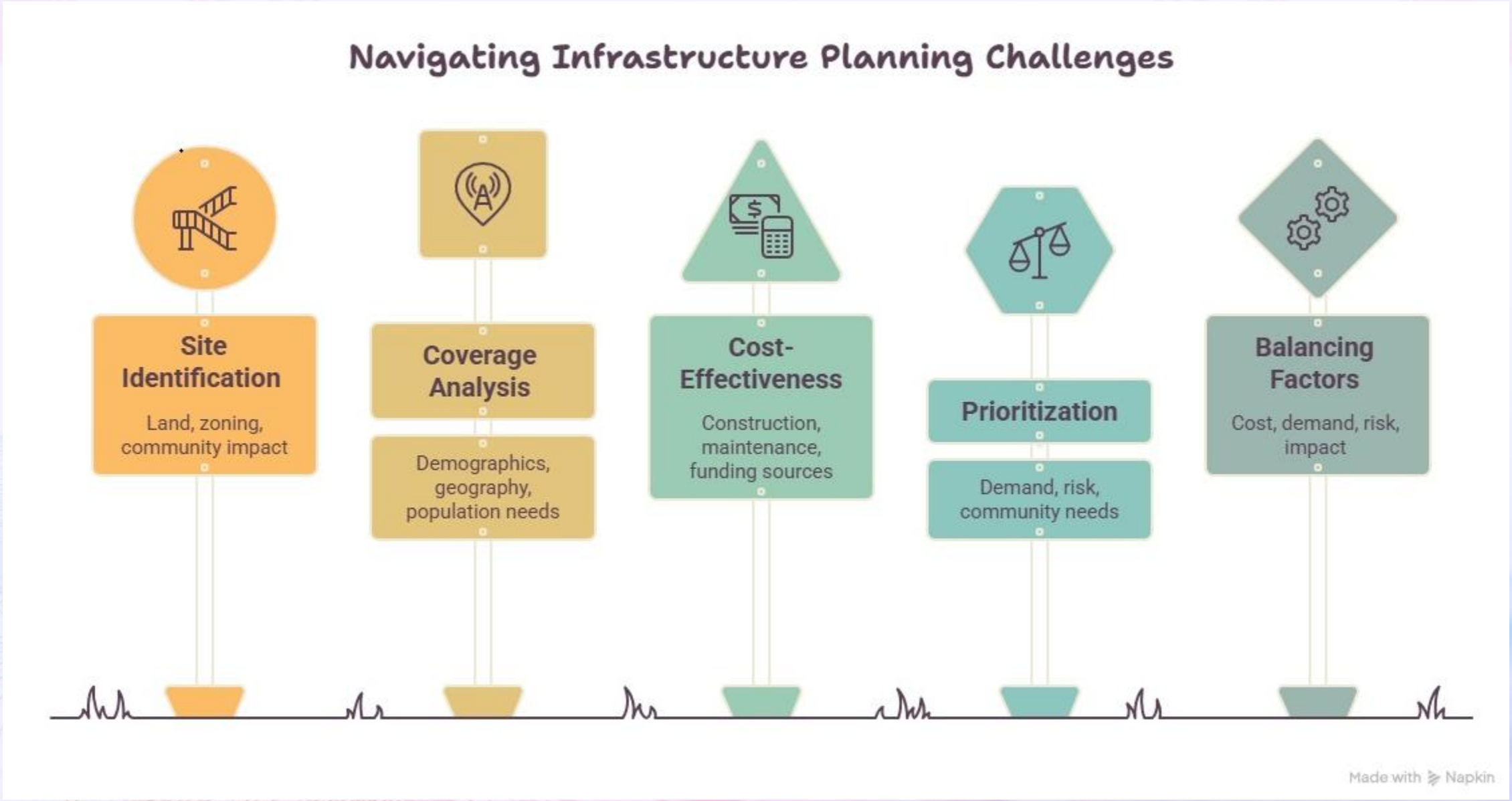
Learning Outcomes

By the end of this session, participants should be able to:

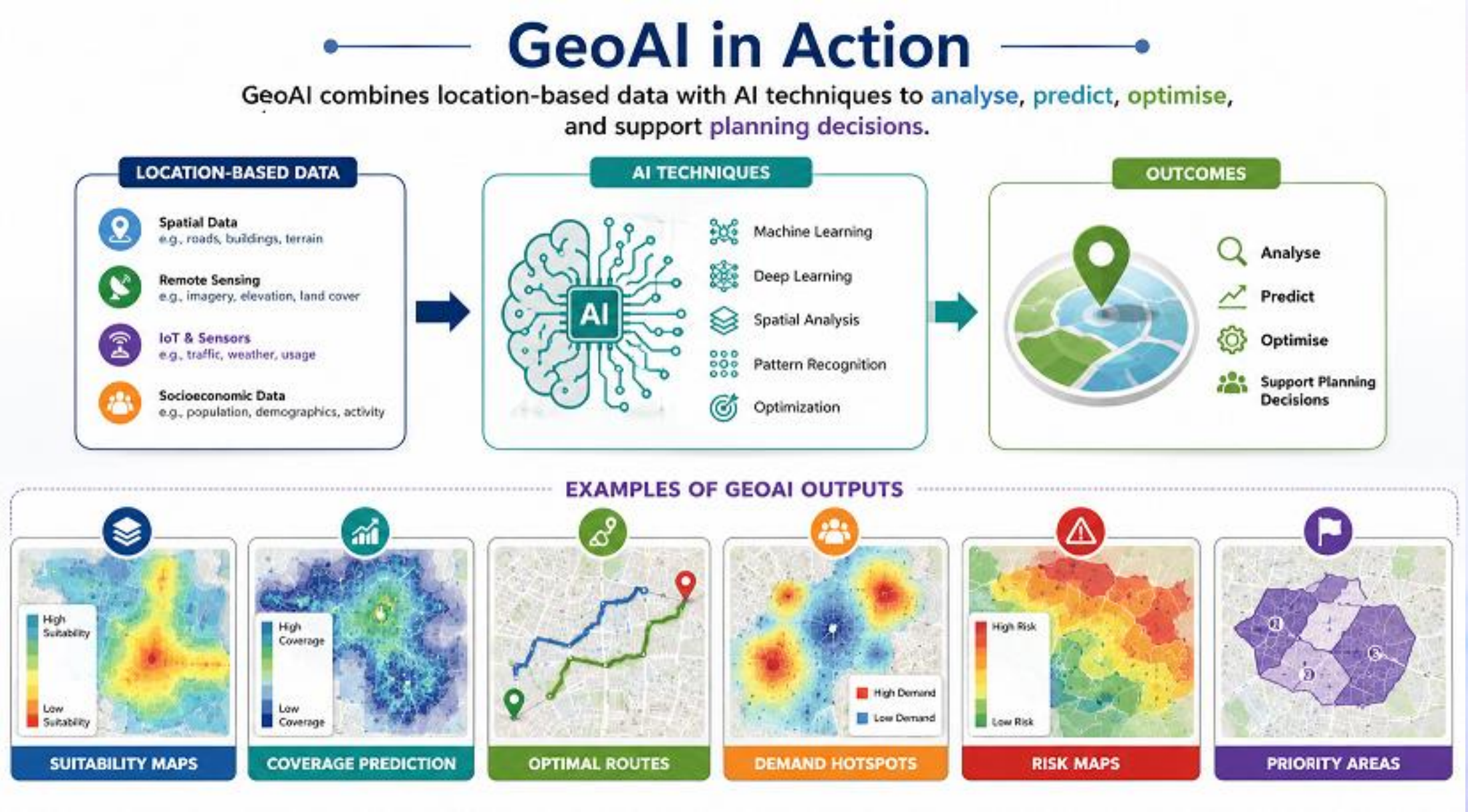
1. Understand how GeoAI supports infrastructure planning
2. Identify key geospatial data used in planning decisions
3. Explain basic applications of GeoAI in site selection, fibre routing, and 5G deployment
4. Recognise how spatial cost-benefit analysis supports better decision-making

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Why Infrastructure Planning Needs GeoAI



Refresh: What is GeoAI?



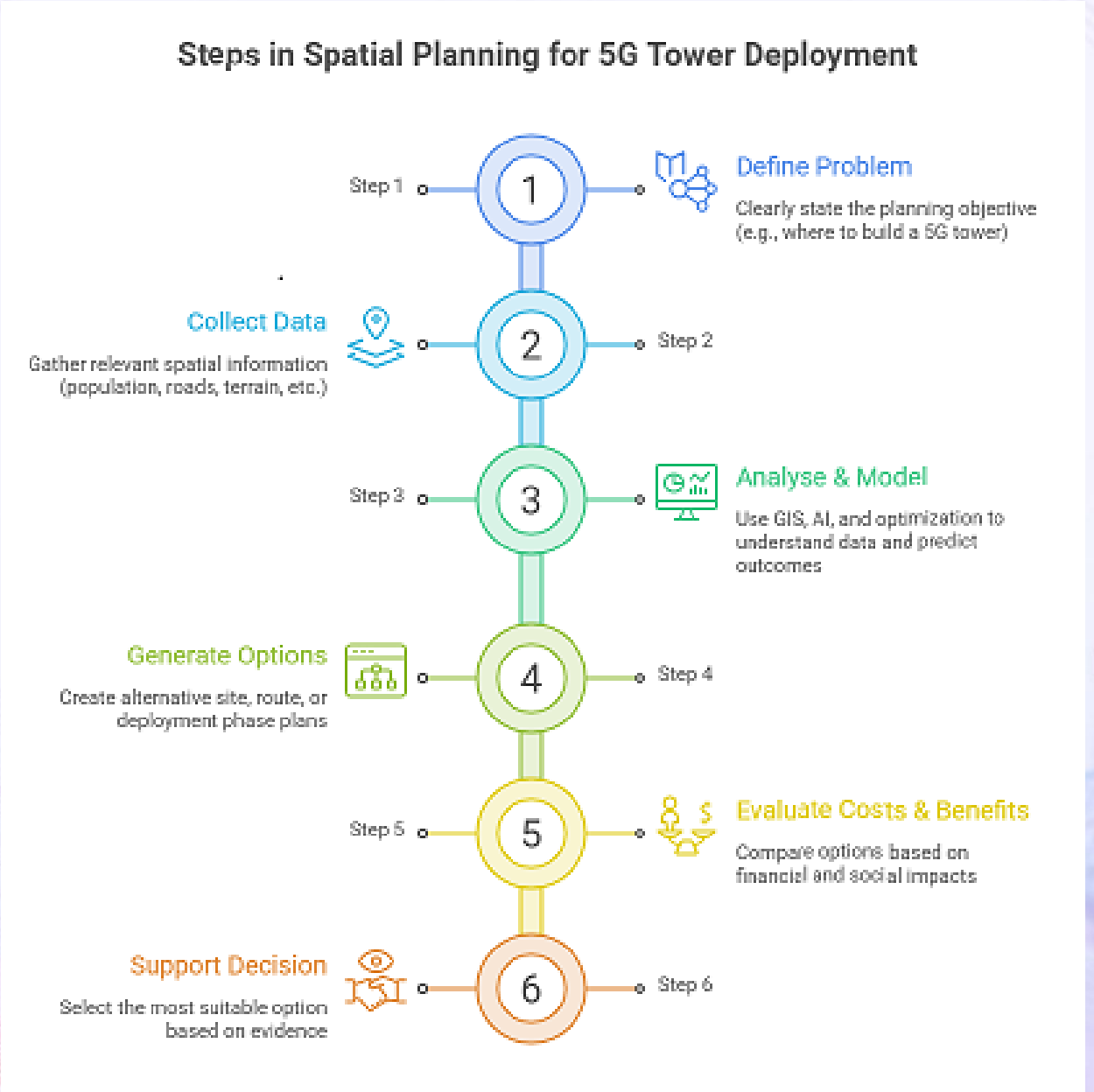
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Types of Data Used in GeoAI Planning



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General GeoAI Planning Workflow




GeoAI in Site Selection
Criteria
High Demand
Good Accessibility
Low Risk
Suitable Land
Near Existing Infrastructure
Reasonable Cost
High Social or Economic Impact
GeoAI in Site Selection
Applications
Telecommunication Towers
Data Centres
Substations
Service Centres
Smart City Facilities
Disaster Response Centres

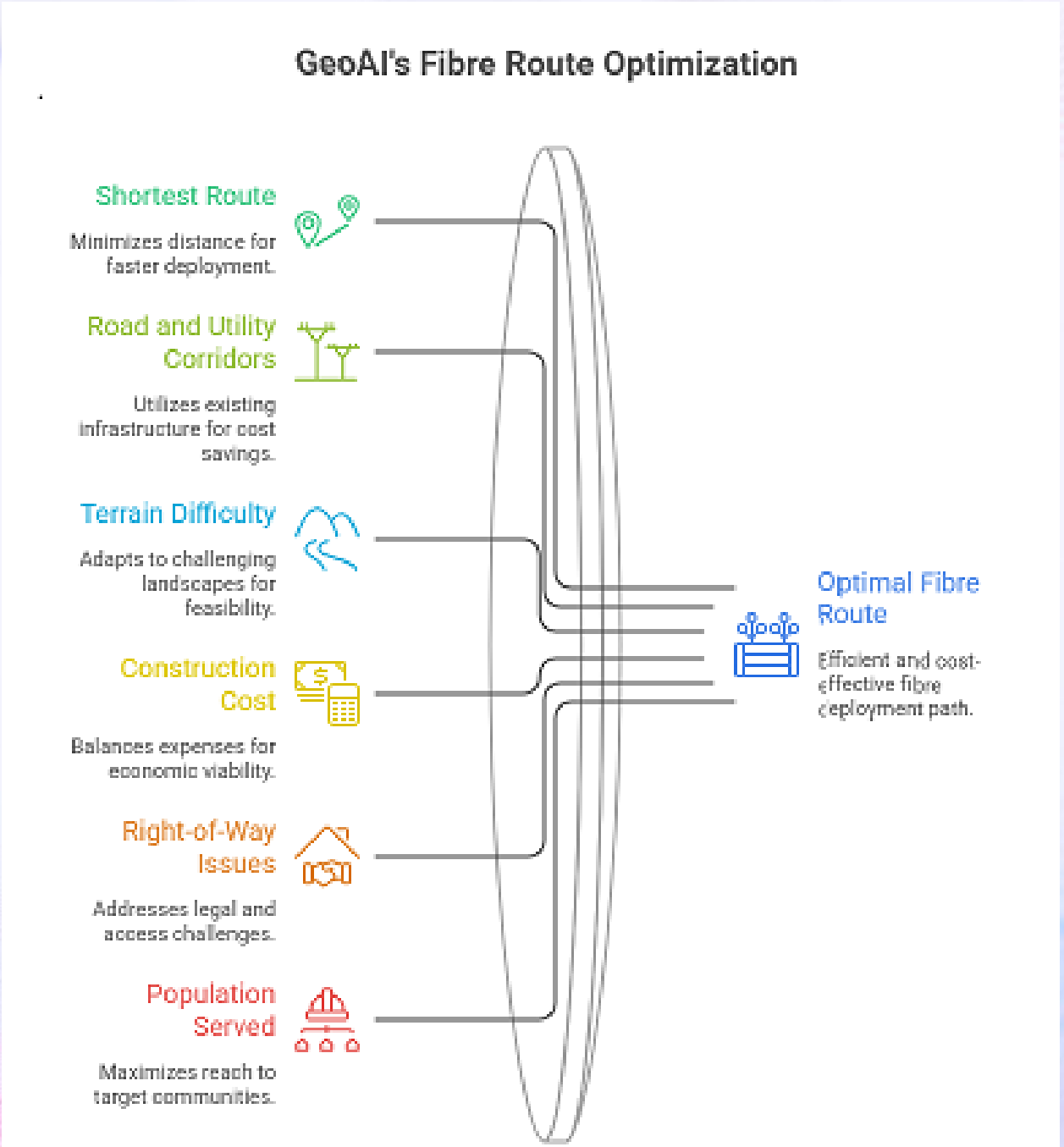
Example — Site Selection Suitability Factors

For example, selecting a new telecom tower location may consider:

Factor	Why It Matters
Population density	Indicates demand
Distance to existing towers	Avoids duplication
Road access	Reduces installation and maintenance difficulty
Elevation	Improves signal coverage
Flood risk	Reduces operational risk
Land availability	Supports practical deployment
Cost	Ensures financial feasibility

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Application 2 — Fibre Route Planning



The shortest route is not always the best route.

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5G DEPLOYMENT STRATEGIES

Data-Driven Planning for Smarter, Faster, Better Coverage

GEOAI INSIGHTS

Coverage Gaps
Identify areas with weak or no 5G signal.

High-Traffic Areas
Detect locations with high user density and demand.

Suitable Tower Locations
Find optimal sites with good coverage and backhaul.

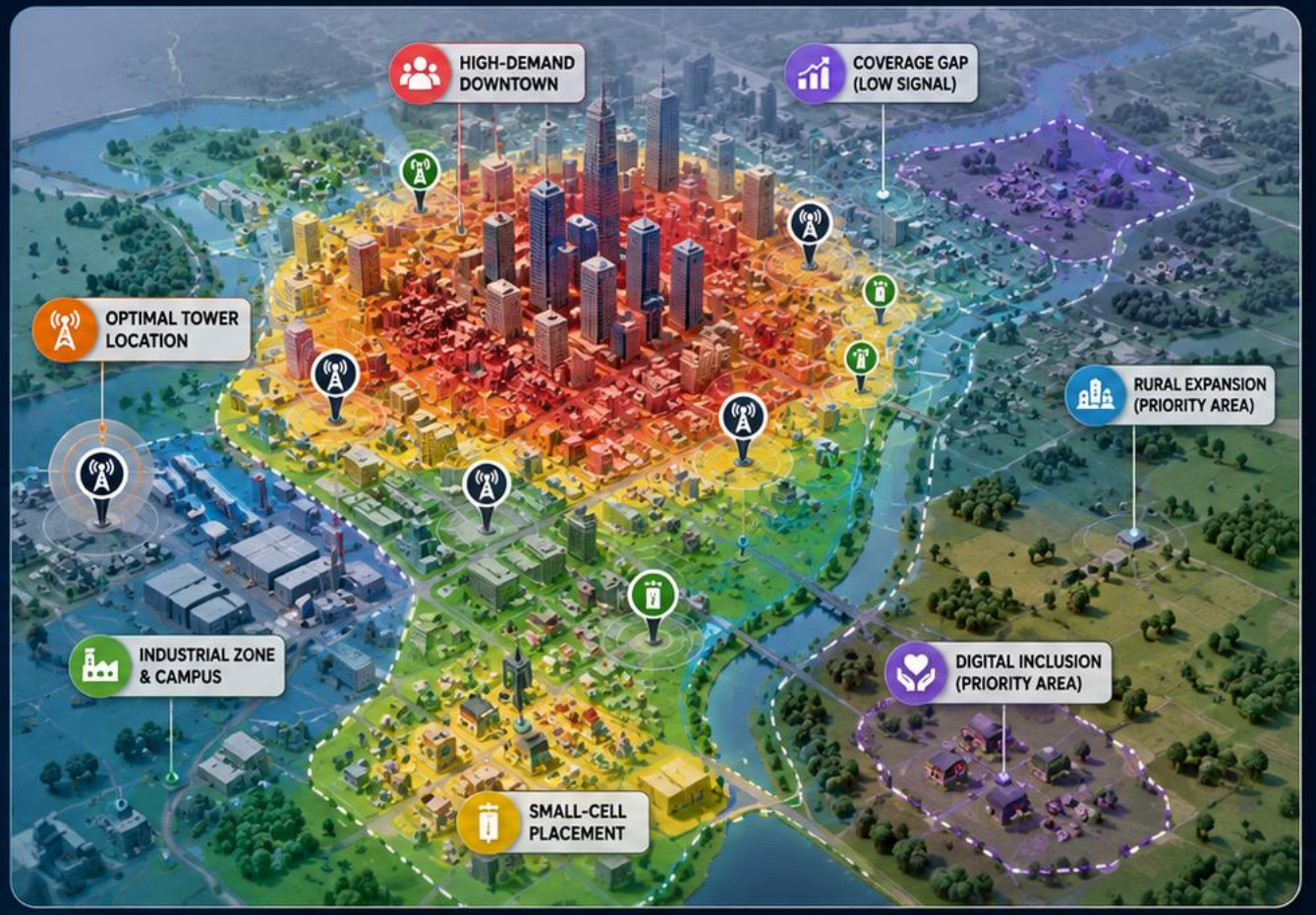
Small-Cell Placement
Determine best spots for small cells in dense areas.

Urban vs. Rural Priorities
Balance investment between urban demand and rural coverage.

Industrial & Smart City Demand
Locate industrial zones, campuses, and smart city use cases.

Expected ROI
Estimate return on investment based on demand and cost.

Digital Inclusion Support
Identify underserved communities that need connectivity the most.



DEPLOYMENT STRATEGY

1. **DEPLOY FIRST IN HIGH-DEMAND AREAS**
Focus on dense urban areas with highest demand and capacity needs.

2. **FILL COVERAGE GAPS**
Extend coverage to underserved areas to ensure reliable service.

3. **SUPPORT INDUSTRIAL ZONES & CAMPUSES**
Enable critical use cases for industries, ports, campuses, and enterprises.

4. **PHASED EXPANSION**
Gradually expand to rural and underserved areas based on priority and ROI.

RESULT: SMARTER 5G INVESTMENT
Better coverage. Higher capacity.
Stronger communities.
Greater returns.

POSSIBLE COSTS

- Equipment cost
- Construction cost
- Maintenance cost
- Land or permit cost
- Operational risk



Compare options based on cost and impact



Choose the option that delivers the greatest impact for the best cost.

POSSIBLE BENEFITS

- Population served
- Improved connectivity
- Economic opportunity
- Service reliability
- Digital inclusion
- Long-term expansion potential

Key Takeaways

- GeoAI helps planners make smarter location-based decisions
- Infrastructure planning should consider demand, cost, risk, and impact
- Site selection, fibre routing, and 5G deployment can be improved using spatial intelligence
- GeoAI supports better, faster, and more transparent planning

GeoAI is not only about technology. It is about making better decisions for people, services, and future infrastructure.

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026